



Sawera Yaseen

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WORK EXPERIENCE

 **SAXION UNIVERSITY OF APPLIED SCIENCES - SMART MECHATRONICS AND ROBOTICS RESEARCH GROUP** – ENSCHEDE, NETHERLANDS

MASTER'S THESIS INTERN – 01/03/2025 – 30/05/2025

- Conducted experiments to improve power line detection using computer vision and sensor fusion techniques for drone-based obstacle avoidance.
- Implemented a stereo vision-based pipeline using YOLOv8 for real-time detection and 3D distance estimation of power lines.
- Developed a monocular vision approach combining UNet-based segmentation with visual-inertial odometry and 2D-3D reconstruction for distance estimation.
- Conducted extensive testing using a ZED2 stereo camera across various distances and scenes to evaluate the feasibility, robustness, and comparative performance of both methods for drone-based obstacle avoidance.

 **NATIONAL CENTER OF ROBOTICS AND AUTOMATION, MUET** – JAMSHORO, PAKISTAN

RESEARCH ASSISTANT – 01/12/2022 – 30/08/2023

- Conducted research on fault diagnosis of rail wheel profiles using deep learning.
- Designed and developed CAD models for rail wheel analysis.
- Performed sensor calibration and image processing for data collection.
- Implemented and trained deep learning models for fault detection and classification.
- Assisted in technical documentation, report writing, and research publications.

SKILLS

Python | C++ | Pytorch, Tensorflow | ROS(robot operating system) | AutoCad | Solidworks | Automation Studio | Proteus Design Suite | LabVIEW | Microsoft Office | MATLAB | Simulink | Raspberry Pi2 | Arduino | LaTeX | Linux | Git | Gazebo | Foxglove Studio

EDUCATION AND TRAINING

09/2023 – 06/2024 Girona, Spain

MASTERS IN INTELLIGENT FIELD ROBOTIC SYSTEMS (SEMESTERS 1 & 2) University of Girona

Autonomous Systems, Multiview Geometry, Machine Learning, Probabilistic Robotics, Robot Manipulation, Hands-on Planning, Hands-on Perception, Hands-on Localization, Hands-on Intervention

Field of study Robotics

10/2024 – 02/2025 Zagreb, Croatia

MASTERS IN INTELLIGENT FIELD ROBOTIC SYSTEMS (SEMESTER 3 SPECIALIZATION) University of Zagreb

Aerial Robotics, Multi-Robot Systems, Deep Learning, Robotic Sensing Perception and Actuation, Human-Robot Interaction, Ethics and Technology

Field of study Aerial Robotics & Multi-Robot Systems

- Thesis Submission, Poster Submission, Thesis Presentation, Poster Presentation, Graduation

Field of study Mechatronics Engineering | **Final grade** 3.95/4.00 |

Thesis Deep Learning Based Smart Spraying System for Disease Control of Vegetables

● **PROJECTS**

Controlling a Swarm of Crazyflies using Reynolds Rules and Consensus Protocol

- Developed a centralized swarm control approach in ROS2 for both simulation and real-world deployment.
- Applied Reynolds Rules for flocking behavior in aerial swarms.
- Implemented the Consensus-Based Rendezvous Protocol for agent convergence.
- Designed consensus-based formation control for geometric swarm configurations.
- Analyzed different communication topologies to evaluate their impact on swarm dynamics.
- Validated swarm control algorithms on real Crazyflies, showcasing effective multi-agent coordination.

Human Detection and Tracking

- Developed a real-time human detection and tracking system using YOLOv9 for accurate object detection.
- Integrated DeepSORT for multi-object tracking, ensuring identity consistency.
- Leveraged Kalman filtering and appearance-based association for robust tracking in dynamic environments.
- Tested the system under various conditions, tackling occlusions, identity switches, and tracking interruptions.

Frontier Based Exploration

- Developed an autonomous exploration system for Kobuki Turtlebot to map unknown environments.
- Integrated frontier-based exploration with RRT-Connect path planning for efficient area coverage.
- Utilized a 2D LiDAR sensor for dynamic mapping, continuously updating the environment map.
- Implemented the Dynamic Window Approach (DWA) for real-time obstacle avoidance and safe navigation.

EKF based SLAM using ArUco Markers

- Developed a Simultaneous Localization and Mapping (SLAM) for Kobuki Turtlebot using a feature-based Extended Kalman Filter (EKF) with ArUco marker range observations.
- Integrated data from wheel encoders, an IMU, and a RealSense camera to enhance localization accuracy.
- Applied dead-reckoning for initial pose estimation and IMU data for orientation correction.
- Used ArUco markers as landmarks for precise mapping and navigation.

Kinematic Control System for a Mobile Manipulator

- Developed a kinematic control system for a mobile manipulator using ROS and Stonefish Simulator.
- Designed an autonomous system for navigation and precise pick-and-place operations using forward and inverse kinematics to compute joint and base velocities.
- Implemented Task-Priority and Damped Least Squares (DLS) methods to optimize multi-tasking capabilities.
- Developed software architecture for navigation, perception, and manipulation of Kobuki Turtlebot.

Deep Learning Based Smart Spraying System Disease Control of Vegetables (Bachelor's Thesis)

- Developed a YOLOv5-based Smart Spraying System integrated with Raspberry Pi, Arduino, and solenoid valves for precise disease detection and pesticide application on vegetable crops.
- Designed and implemented a mobile platform powered by DC motors and solar energy to identify diseased plants and apply pesticides accurately, reducing chemical waste.

● **HONOURS AND AWARDS**

Erasmus Mundus Joint Master Degree (EMJMD) Scholarship

Awarded the Erasmus Mundus scholarship for the Intelligent Field Robotic Systems (IFRoS) program.

Merit Scholarship – Mehran University of Engineering and Technology

Awarded a scholarship for consistently high academic performance during Bachelor’s in Mechatronics Engineering.

Silver Medal – Mehran University of Engineering and Technology

Recognition for outstanding performance and achievement in the department during the Bachelor's degree.

Research Funding – Ignite National Technology Fund

Received funding for conducting research on "Deep Learning Based Smart Spraying System for Disease Control of Vegetables" as part of the final year project for my Bachelor's degree.

● **ACTIVITIES**

07/2024 – 09/2024
Summer Vacation

(Joensuu, Finland), (Barcelona, Spain), (Jamshoro, Pakistan)

21/06/2025 – 16/07/2025
PhD Application & Interview Preparation

Engaged in proposal development, selection tasks, and interviews for doctoral program.

17/07/2025 – 30/08/2025
Summer Vacation

Jamshoro, Pakistan